FACULTY OF ENGINEERING TECHNOLOGY

Subject: [redacted] Mid Exam Fall 24/25

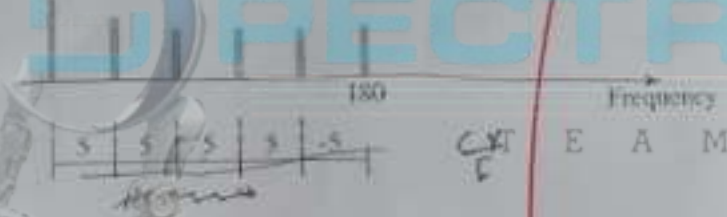
Name: [redacted] Duration: 50 min

THIS EXAM CONTAINS TWO QUESTIONS

Q1:(Please write your answers into table only, otherwise your answers will be neglected) (24 marks)

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	B	A	B	A	B	B	B	A	C
Question	11	12								
Answer	C	D								

1. What is the bandwidth of the composite signal shown in figure:



a) 175 b) 25 c) 180 d) 155

2. A signal has passed through three cascaded amplifiers, each with a 4 dB gain. How much is the signal amplified?

a) 15.85 b) 2.5 c) 15 dB d)12

3. A line has a signal-to-noise ratio of 1000 and a bandwidth of 4 KHz. What is the maximum data rate supported by this line?

a) ≈ 40bps b) ≈ 40kbps c) ≈ 20bps d) ≈20 kbps

4. A computer monitor has a resolution of 1200 by 1000 pixels. If each pixel uses 1024 colors, and each pixel is represented by 10 bits, how many bits are needed to send the complete contents of a screen?

a) 12,000,000 bits b) 12.288 Gbits

c) 1.2288 Gbits d)122 kbit

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5. What is the transmission time of a packet sent by a station if the length of the packet is 1 million bytes and the bandwidth of the channel is 200 Kbps?

a) 5 s b)40 s c) 200 ms d) 50 ms

6. To deliver the message from the source host to the destination host, is the duty of:

a) MAN b) LAN c) WAN

7. To deliver the message from the source host to the destination host, is the duty of:

a) application layer b) network layer
c) transportation layer d) application layer

8. In a communication system, the noise is most likely to affect the signal:

a) at the channel b) at the transmitter
c) at the destination d) not affected

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9. The minimum problem with traffic is found in:

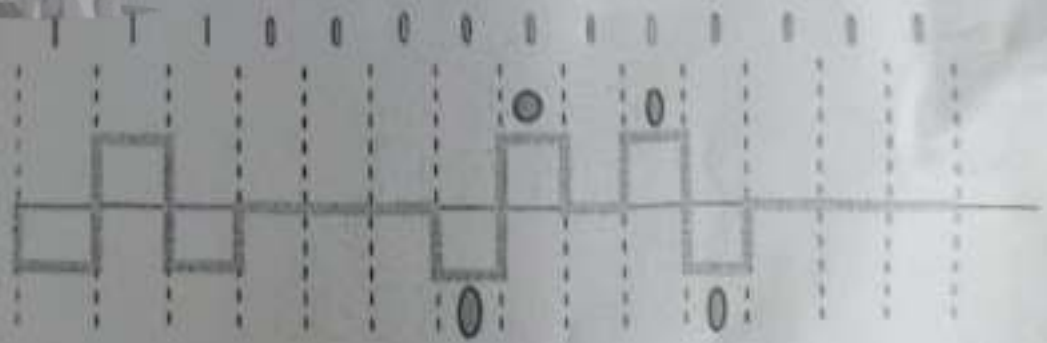
a) star topology b) mesh topology
c) bus topology d) line topology

10. How many invalid (unused) code sequences can we have in 3B/4B encoding?

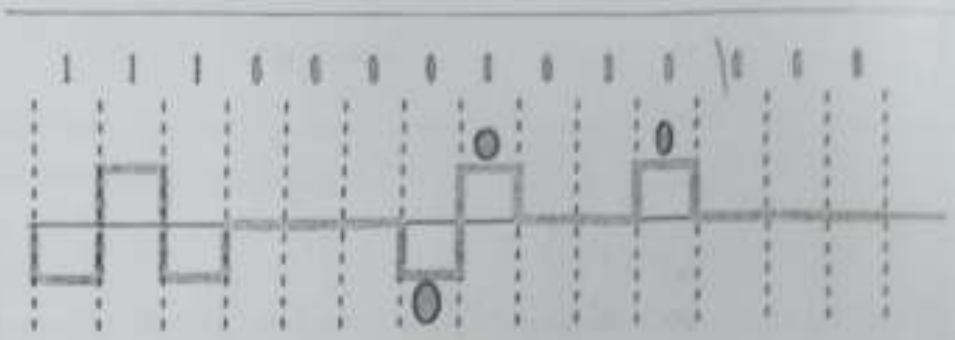
a) 16 b) 32 c) 8 d) 4

11. What is the result of scrambling the sequence 1110000000000 using B8ZS technique? Assume that the last non-zero signal level has been positive.

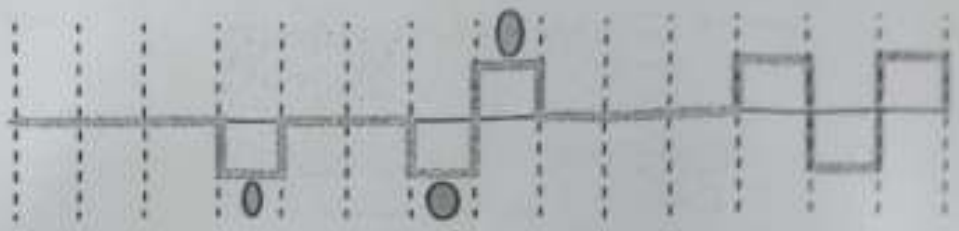
a)



b)



c)



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12. We have a baseband channel with a 1-MHz bandwidth. What is the data rate for this channel if we use Manchester coding scheme?
a) 1 Mbps b) 2 Mbps c) 3 Mbps d) 4 Mbps

Q2. Suppose that the spectrum of a channel is between 3MHz and 4 MHz and SNR_{dB} = 24 dB. Calculate the capacity of the channel. (6 marks)

$C = B \times Log_2(1 + SNR)$

$C = B(F_H - F_L) Log_2(1 + 24)$

$C = (4 - 3) Log_2(1 + 24)$

$C = 1 \otimes Log_2(25)$

GOOD LUCK

$1 \otimes 4.643$

$= 4.643$

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